



Zillow Model for Finding the Best Properties to Flip in Chapel Hill

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Property Flippers Struggle to Find Undervalued Homes with Potential Upside

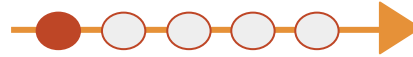
Stakeholder: Chapel Hill Property Flippers

Pain Points: Identifying the right property

Opportunity: Rising demand for house flippers in North Carolina University Towns



Finding a flip can take up to **20%** of the 4-6 month average project timeline



9.2% of North Carolina houses are flipped (8.7% national average)



Lack of data-driven house selection increases risk of failure



1,200 properties flipped in the Triangle per year



Sources: [Ankney](#), [Fox8](#), [Duncan](#), [Mehta](#)

What are the best houses to flip in Chapel Hill that are undervalued and have strong resale potential?

Defining a flip:

Any home that is bought, renovated, and sold within **12 months** qualifies as a **flipped property**

Current Approach:

Flippers search for low listing prices, older homes, “Fixer-upper” tags, and high value neighborhoods **OR** pay an agent to do these steps

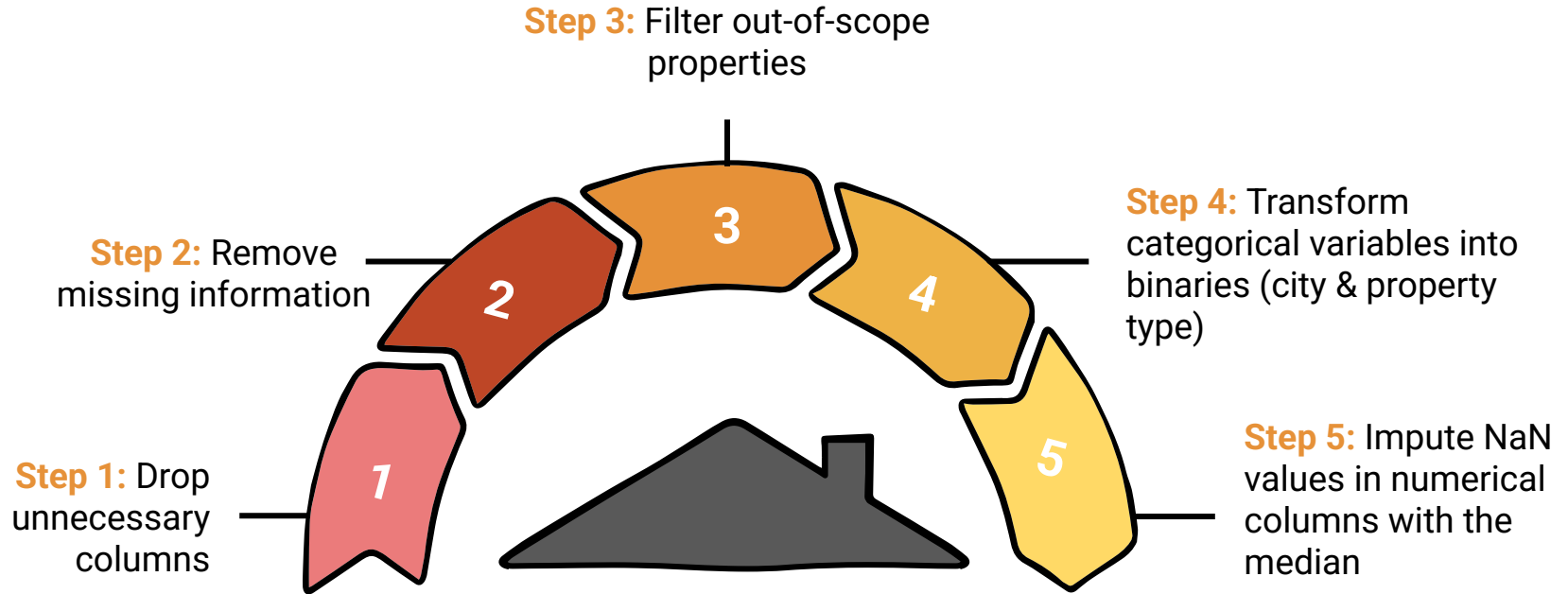
Issues:

Time consuming, high chance of missing undervalued properties, and difficulty determining accurate buyer interest and resale potential.

Our Solution: Model that identifies best properties to flip

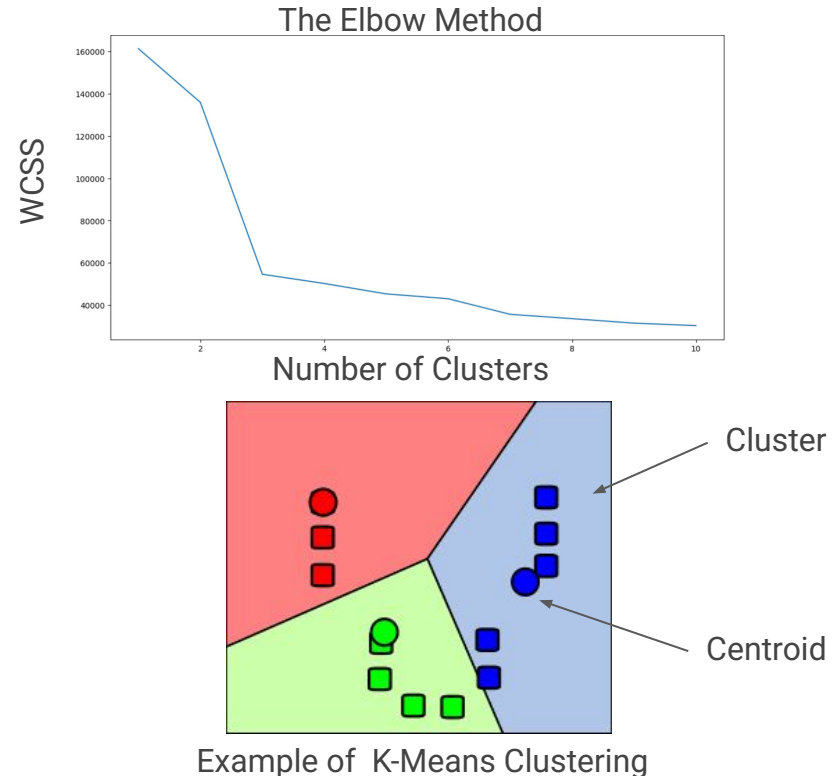
Sources: [Investopedia](#), [FlippingHouses2025](#)

Wrangled and Cleaned our 2025 Zillow Dataset



Finding Undervalued Homes: Methods + Approach

- **K-Means Clustering:** Unsupervised machine learning algorithm used to group data into **k distinct clusters** based on feature similarity.
- **The Elbow Method:** For a range of k values 1-10 calculate WCSS (Within-Cluster Sum of Squares) which measures how tightly the data points are clustered around the centroids. We look for the “**elbow**” point in the graph. This is where the WCSS starts to decrease more slowly. This is the point of optimal number of clusters
- **Features:** `'longitude'`, `'latitude'`, `'livingAreaValue'`, `'yearBuilt'`, `'bedrooms'`, `'bathrooms'`, `'single_family'`, `'chapel_hill'`



Our Model Outputs 57 Flippable Properties

The Model

- clusters properties based on similarity
- calculates the median price of each cluster
- creates an indicator value ('**flippable**') of how many standard deviations below the median price a house is (0 if above)
 - possibility of flipping a house

The 70% Rule

"Don't pay more than 70% of the property's after-repair value (ARV) minus the cost of repairs"

$$\text{price} = 0.7\text{ARV} - \text{cost}$$

											flippable
zpid	price	zestimate	livingAreaValue	address	single_family	chapel_hill	arv	raw_profit	renovation_cost		
17675	88951629.0	375000	723450.0	2166.0	200 Eastwood Rd	1	1	614776	239776	55343.2	
19727	71844443.0	625000	641800.0	3460.0	430 Smith Level Rd	1	1	1092261	467261	139582.7	
23609	50032016.0	400000	405000.0	1820.0	105 Mary St	1	0	587574	187574	11301.8	

Source: [Forbes](#)

The Model



15,347
Unflippable

vs

669
Flippable



Used K-Fold Stratification

Precision > Recall

Flipping a bad house
(**false positive**) is more
costly than missing a
good one (**false
negative**)

Confusion Matrix

15,006	48
415	241

Our Model Accuracy: 0.9700

Accuracy if only outputted False:
0.9582

Impact: Minimizing Flip Risk with 97% Accuracy

Time Saved

Significant decrease in time spent searching for properties: from weeks to minutes



Reduced Risk

Finds statistically undervalued homes with high resale potential rather than relying on manual search

Optimized ROI Potential

Top 3 homes pass the 70% rule with profits of:

- \$239,776
- \$467,261
- \$187,574

Mission Statement: Commitment to providing real solutions for homeowners who are ready to move and serving both communities and neighborhoods

Moving Forward:

Apply the same data pipeline process to **future Chapel Hill** datasets or datasets in **similarly promising cities**

Sources: [TLC Home Buyers](#), [Property Checker](#), [New Western](#)



Thank you!